Sales Data Analysis Report (Project 2)

1. Introduction

This is my second data analysis project, where I worked on a sales dataset to extract meaningful insights and improve my analytical thinking. The aim of this project is to understand sales patterns, product performance, and regional trends using data analysis techniques.

2. Objectives

- Analyze product-wise performance
- Understand city-wise sales distribution
- Identify trends and patterns in sales
- Detect high and low-performing products
- Generate actionable business insights

3. Dataset Overview

The dataset contains transactional sales data with the following key features:

- Product Name
- City
- Quantity
- Total Revenue (Total Price)
- Transaction-level details

This dataset helps in understanding both customer purchasing behavior and business performance.

4. Data Cleaning & Preprocessing

Before analysis, the dataset was cleaned to ensure accuracy and consistency:

- Handled missing values
- Removed duplicate records
- Converted data types (numeric and categorical)
- Standardized column values for consistency

5. Exploratory Data Analysis (EDA)

5.1 Product Performance Analysis

Analyzed revenue generated by each product to identify top and low performers.

Insight:
A small number of products contribute significantly to total revenue, indicating high demand and business importance.

5.2 City-wise Sales Analysis

Compared total revenue across different cities.

Insight:
Certain cities generate higher revenue, suggesting strong market presence and potential for expansion.

5.3 Sales Trend Analysis

Analyzed how sales change over time (if date data is available).

Insight:
Sales patterns may show seasonal trends, which can help in planning inventory and marketing strategies.

5.4 Revenue Distribution

Studied how revenue is distributed across products.

Insight:
A large portion of revenue comes from a limited number of products (Pareto-like distribution).

5.5 Order Behavior Analysis

Analyzed quantity distribution across cities.

Insight:
Most cities show a higher frequency of low-quantity orders, indicating a preference for smaller purchases.

5.6 Low Performing Products

Identified products with minimal contribution to revenue.

Insight:
These products may require pricing adjustments, better marketing, or removal.

6. Visualizations Used

- Bar charts for product and city analysis
- Line charts for trend analysis
- Pie/Donut charts for revenue distribution
- Heatmap for city vs quantity distribution

7. Key Business Insights

- Top-performing products drive the majority of revenue
- High-revenue cities should be prioritized for growth
- Seasonal patterns can guide business planning
- Low-performing products need strategic improvement
- Customer behavior indicates preference for smaller orders

8. Conclusion

This project helped in understanding how raw data can be transformed into meaningful insights. By analyzing sales data, important patterns and opportunities were identified that can support better business decisions.

As my second project, this work reflects improvement in both technical skills and analytical thinking.3

9. Future Scope

- Implement sales forecasting using machine learning
- Perform customer segmentation analysis
- Build an interactive dashboard using Power BI or Tableau
- Apply advanced statistical techniques for deeper insights

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